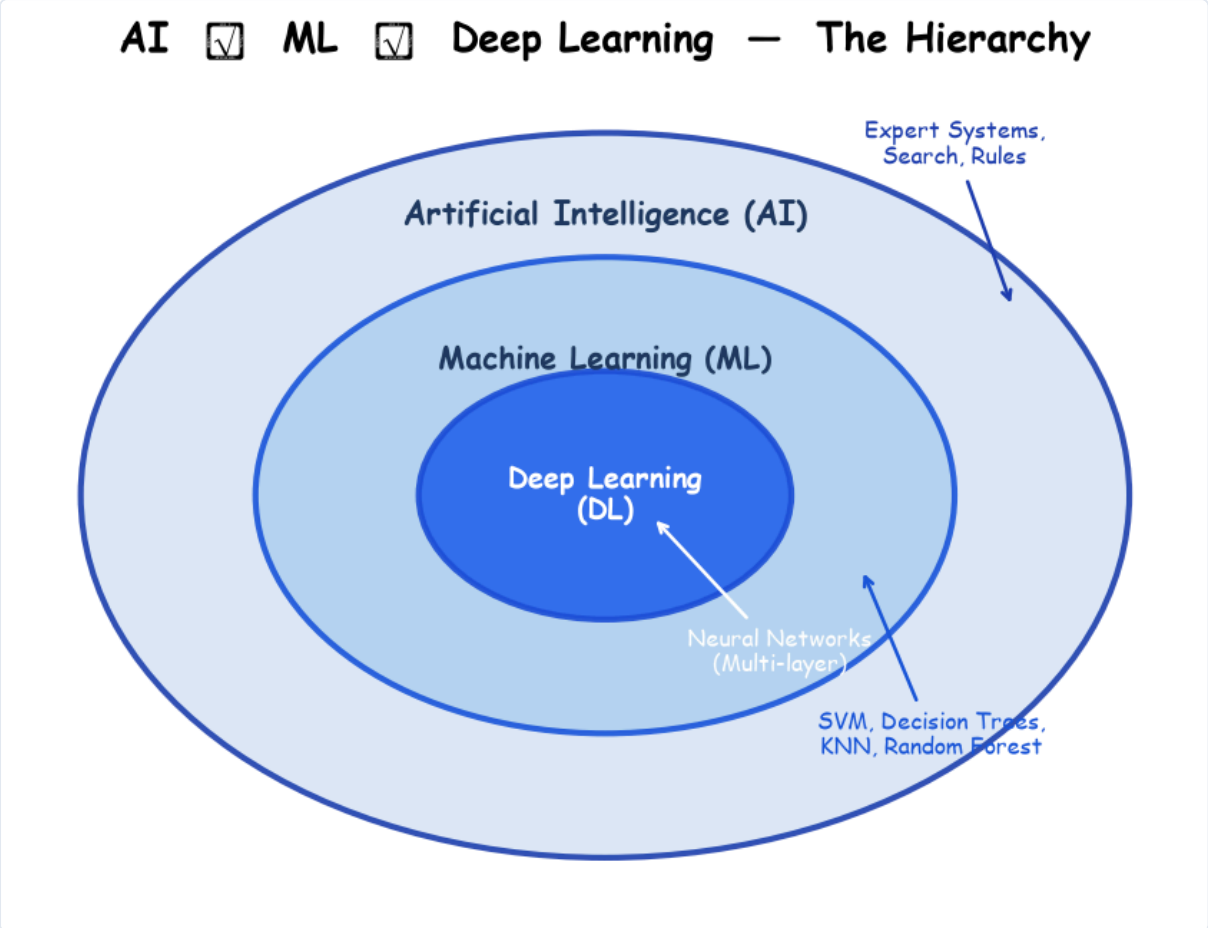
Submodule 36.01: Deep Learning kya hai? (Hinglish)

1. Deep Learning ki Paribhasha

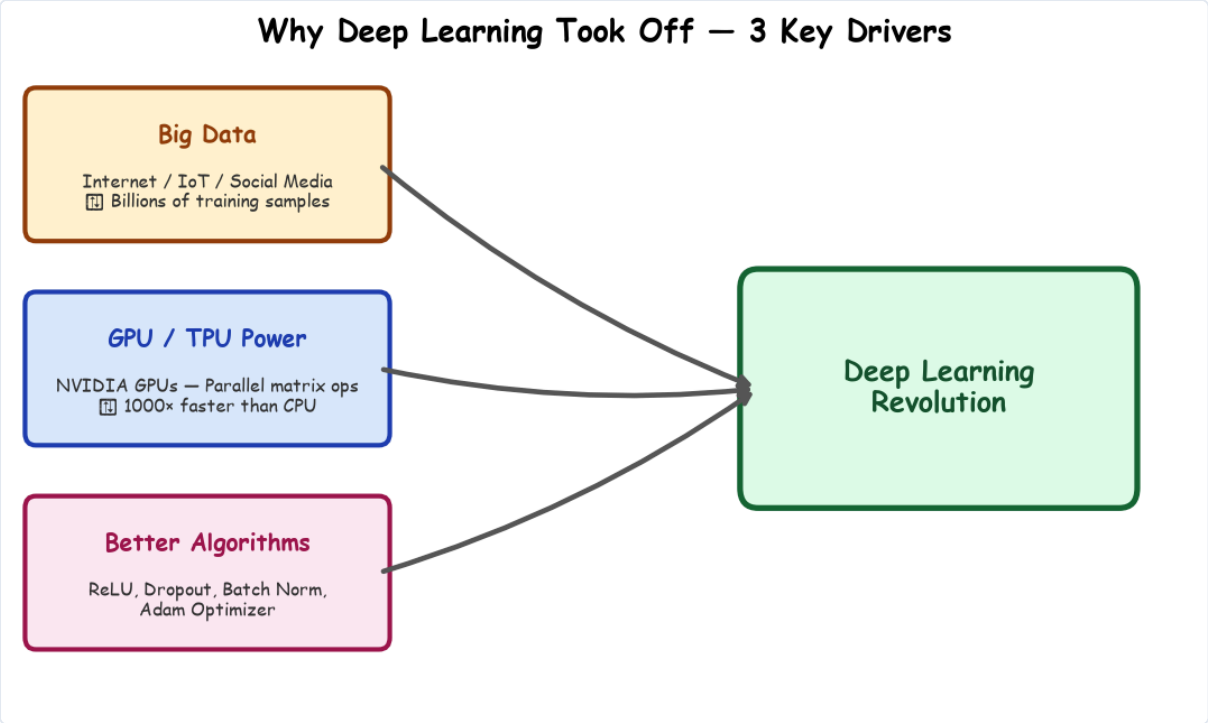
Deep Learning (DL) dar-asal **Machine Learning (ML)** ka ek advanced subset hai, jo khud **Artificial Intelligence (AI)** ke andar aata hai. Deep Learning human brain (insani dimaag) ke neurons aur neural networks ke structure se inspire hokar banaya gaya hai.



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2. Deep Learning Achanak Itna Popular Kyun Hua?

Neural networks ki theory 1950s mein hi invent ho chuki thi, lekin pichle 10-15 saalon mein Deep Learning ne poori duniya badal di. Iske piche teen bade factors hain:



1. **Big Data ki Uplabdhta**: Internet aur smartphones ki wajah se aaj hamare paas billions of images, videos, audios aur text documents maujood hain.
2. **GPU aur Compute Power**: Neural network mein lakho-crodon matrix multiplications hote hain. NVIDIA ke modern GPUs in calculations ko CPU ke mukable hazaron guna fast process kar sakte hain.
3. **Advanced Algorithms**: Naye activation functions (jaise **ReLU**) aur optimizers (jaise Adam) ne puraani training problems ko solve kar diya.

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3. Real-World Applications

Field	Real-Life Example	Deep Learning Technology
Computer Vision	Self-driving cars (Tesla), Medical X-ray aur MRI scan analysis	Convolutional Neural Networks (CNN)
Natural Language Processing	ChatGPT, Google Translate, AI Chatbots	Transformers aur Attention Models
Voice & Speech	Siri, Alexa, Speech-to-text transcription	Recurrent Neural Networks (RNN)
Recommendation Engines	YouTube videos aur Netflix movies ki recommendation	Deep Neural Collaborative Filtering

Field	Real-Life Example	Deep Learning Technology
Generative AI	AI image generation (Midjourney), AI video generation	Diffusion Models aur GANs